

2	Concepts tested: Work, Energy, Power	
(a)	For an object mass m , moved by a force F , undergoing acceleration a through a displacement s , with initial velocity u and final velocity v . $F = ma$ Work done = $Fs = mas$	M1
	$v^2 = u^2 + 2as$	M1
	$\text{Work done} = mas = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \text{Gain in kinetic energy}$ if $u = 0$, work done = kinetic energy	M1
	$E_k = \frac{1}{2}mv^2$	A0
(b)(i)	$\text{EPE} = \frac{1}{2}ks^2 = \frac{1}{2}(120)(10 \times 10^{-2})^2 = 0.60 \text{ J}$	A1
(b)(ii)	Using COE: $\text{KE}_i + \text{GPE}_i + \text{EPE}_i = \text{KE}_f + \text{GPE}_f + \text{EPE}_f$ $0 + (2.0)(9.81)(D \sin 40) + 0 = 0 + 0 + \frac{1}{2}(120)D^2$	C1
	$D = 0.210 \text{ m}$	A1
(b)(iii)	D will increase.	B1
(b)(iv)	Using COE: $\text{KE} + \text{GPE} + \text{EPE} = \text{KE} + \text{GPE} + \text{EPE} + W_{\text{by box}}$ $0 + (2.0)(9.81)(D' \sin 40) + 0 = 0 + 0 + \frac{1}{2}(120)D'^2 + (5.0)D'$	C1
	$D' = 0.127 \text{ m}$	A1

3 (a)	Simple harmonic motion is defined as the motion of an object whose	B1
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